

Beyond Brute-Force Pretraining

A Meta-Analysis of Three Recent Approaches to Building More Capable Large Language Models

1. Introduction

Large language models (LLMs) such as GPT-4, PaLM, LLaMA, and Gemini have become the dominant paradigm in natural language processing, achieving near-human performance across reasoning, coding, dialogue, and knowledge-intensive tasks. Yet the cost of building these systems — in compute, data, energy, and human feedback — is growing faster than the resulting capability gains. As a result, a substantial portion of recent research has shifted away from simply scaling pretraining and toward smarter ways of *combining*, *surveying*, and *refining* what already exists.

This report performs a meta-analysis of three recent works that each tackle this challenge from a different angle. Together they trace a coherent arc: one paper proposes a concrete technique for fusing the knowledge of multiple pretrained LLMs into a single stronger model; one paper provides a panoramic survey that maps the landscape of architectures, training strategies, and evaluation methods; and one paper focuses on the alignment stage, asking how human feedback can be collected more efficiently. The unifying theme is **improving LLM capability without proportional growth in cost** — whether the cost is measured in pretraining FLOPs, redundant model weights, or human labelling effort.

Papers analyzed:

- **Wan et al. (2024).** *Knowledge Fusion of Large Language Models*. Published at ICLR 2024.
- **Naveed et al. (2024).** *A Comprehensive Overview of Large Language Models*. arXiv:2307.06435v10 (preprint, Oct 2024).
- **Dwaracherla, Asghari, Hao, and Van Roy (2024).** *Efficient Exploration for LLMs*. Proceedings of the 41st International Conference on Machine Learning (ICML 2024), PMLR 235.

2. Paper Summaries

2.1 FuseLLM — Knowledge Fusion of Large Language Models

Wan, F., Huang, X., Cai, D., Quan, X., Bi, W., & Shi, S. (2024). *Knowledge Fusion of Large Language Models*. ICLR 2024.

Problem. Training LLMs from scratch is expensive and produces redundant capabilities across organisations. Traditional ways of combining models — ensembling or weight merging — either require running every model at inference time or assume identical architectures, neither of which is realistic for modern open-source LLMs.

Method. The authors introduce FuseLLM, which externalises the knowledge of multiple source LLMs through their *generative probability distributions* over text. For each training token, the probability vectors produced by each source model are aligned (using a minimum-edit-distance token alignment strategy that improves on prior exact-match approaches) and then fused. Two fusion functions are studied: **MinCE**, which selects the distribution with the lowest cross-entropy against the gold label, and **AvgCE**, a cross-entropy-weighted average. The target LLM is then continually trained with a weighted combination of standard causal language-modeling loss and a divergence-minimisation loss against the fused distribution.

Setup & results. Three 7B open-source models with distinct architectures — Llama-2, OpenLLaMA, and MPT — were fused into a single Llama-2 target. Training used MiniPile (~1.8B tokens, <0.1% of Llama-2's pretraining corpus). Evaluations spanned 42 tasks across Big-Bench Hard (reasoning), Common Sense (zero-shot multiple choice), and MultiPL-E (code generation in 10 languages). FuseLLM achieved relative improvements of 5.16% on BBH, 1.25% on CommonSense, and 6.36% on MultiPL-E over Llama-2, outperforming continual pretraining and conventional knowledge distillation from a larger 13B teacher. MinCE consistently beat AvgCE, suggesting that preserving the sharpness of the best individual model is more valuable than averaging.

2.2 A Comprehensive Overview of Large Language Models

Naveed, H., Khan, A. U., Qiu, S., Saqib, M., Anwar, S., Usman, M., Akhtar, N., Barnes, N., & Mian, A. (2024). *A Comprehensive Overview of Large Language Models*. arXiv preprint arXiv:2307.06435v10.

Problem. The LLM literature has expanded explosively — from roughly 60 papers in 2019 to over 28,000 in 2024 — making it nearly impossible for newcomers and even specialists to maintain a coherent view of the field. This paper is a survey rather than a novel method; its contribution is structural and synthetic.

Scope. The authors organise the field into seven branches: pre-training, fine-tuning, efficiency, inference, evaluation, applications, and challenges. They provide structured summaries of roughly forty pre-trained LLMs (T5, GPT-3, PaLM, LLaMA-1/2/3, Mixtral, DeepSeek-v2, Gemini, Nemotron-4, and many others) and dozens of instruction-tuned

and aligned variants, covering architecture details (attention type, positional encoding, normalisation, activation), training hyperparameters, data preprocessing, and observed insights such as the empirical scaling laws of Chinchilla.

Key contributions. Beyond cataloging, the survey distils cross-cutting lessons: (i) decoder-only architectures dominate, though encoder-decoder variants with mode-switching tokens (UL2, AlexaTM) remain competitive; (ii) parameter-efficient fine-tuning methods (LoRA, adapters, prefix tuning, BitFit) and post-training quantisation (SmoothQuant, GPTQ, QLoRA) have made deployment of 70B+ models feasible on modest hardware; (iii) alignment via RLHF, DPO, RAFT, and constitutional AI is now standard practice; (iv) retrieval-augmented and tool-augmented LLMs allow smaller models to match the performance of much larger ones; and (v) hallucinations, bias, privacy leakage, and adversarial robustness remain unsolved.

2.3 Efficient Exploration for LLMs

Dwaracherla, V., Asghari, S. M., Hao, B., & Van Roy, B. (2024). *Efficient Exploration for LLMs. Proceedings of the 41st International Conference on Machine Learning, PMLR 235.*

Problem. Reinforcement learning from human feedback (RLHF) typically samples response pairs *passively* from the base LLM and asks raters which is better. The authors ask whether the queries themselves should be chosen more intelligently — that is, whether *active exploration* can reach the same level of alignment using far fewer human preference labels.

Method. The authors compare four query-selection strategies in a controlled pipeline where Gemini Nano generates candidate responses and a larger Gemini Pro acts as a simulated human rater: **passive** sampling, **Boltzmann** exploration (temperature-weighted selection over a point-estimate reward model), **infomax** (maximising the predictive variance of the reward), and **double Thompson sampling** (independently sampling two epistemic indices and choosing the argmax of each). The latter two require uncertainty estimates, which are supplied by an epistemic neural network (ENN) reward model — an ensemble of MLP heads placed on top of the frozen Nano embedding torso.

Results. Active exploration produces large gains. Double Thompson sampling reaches the win-rate of passive exploration with roughly an order of magnitude fewer queries (e.g. the performance attained by ~30,000 passive queries is matched by ~3,000 double-TS queries). Boltzmann improves over passive, infomax improves further, but double TS dominates, reaching win rates near 0.68. Crucially, the gap between Boltzmann and double TS shows that *uncertainty estimates*, not just active selection, are what drives the improvement. The ENN reward model achieves substantially better dyadic joint negative-log-loss than a point estimate while maintaining equivalent marginal NLL, confirming that its uncertainty estimates are meaningful.

3. Comparative Analysis

Although the three papers differ in genre — a method paper, a survey, and an empirical exploration study — they can be productively compared along several axes. The table below summarises the comparison.

Dimension	FuseLLM (Wan et al.)	LLM Overview (Naveed et al.)	Efficient Exploration (Dwaracherla et al.)
Genre	Method paper proposing a novel fusion technique	Survey / structural reference	Empirical study of alignment-stage algorithms
Primary objective	Combine the strengths of multiple pretrained LLMs into one target model	Map the entire LLM research landscape and distil cross-cutting insights	Reduce the number of human preference labels needed for RLHF
Stage of LLM lifecycle	Continual pretraining	All stages (pretraining → fine-tuning → deployment)	Reward modelling / preference data collection
Core technique	Token-aligned distribution fusion with MinCE/AvgCE; lightweight continual training	Synthesis & taxonomy; no new technique introduced	Double Thompson sampling with an ENN reward model
Models used	Llama-2 7B, OpenLLaMA 7B, MPT 7B (source); Llama-2 7B (target)	Reviews ~40 LLMs spanning 11B–1.085T parameters	Gemini Nano (agent) and Gemini Pro (simulated rater)
Data	MiniPile (~1.8B tokens, <0.1% of Llama-2 pretraining)	N/A — surveys datasets used by others (C4, Pile, ROOTs, etc.)	Anthropic Helpfulness base prompts; ~40,000 synthetic preference queries
Evaluation	BBH (27 tasks), CommonSense (5 tasks), MultiPL-E (10 languages); also perplexity vs. ensemble & weight merging	Reviews dozens of benchmarks (MMLU, SuperGLUE, BIG-bench, HumanEval, GSM8K, TruthfulQA, etc.)	Win-rate against Gemini Nano on held-out Anthropic prompts; dyadic joint NLL for uncertainty quality
Headline result	+5.16% on BBH, +1.25% CS, +6.36% ME over Llama-2; beats KD from Llama-2 13B	Identifies decoder-only + RLHF + PEFT + RAG as the dominant recipe	~10× reduction in queries needed to reach a target win-rate
Key strength	Architecture-agnostic; no inference overhead at deployment; modest training cost	Breadth and synthesis; useful as an onboarding reference	Demonstrates that exploration quality, not just data volume, drives RLHF gains
Key limitation	Performance on coding still trails the best source model; sensitive to training corpus relevance	As a static survey, becomes outdated quickly; depth varies by topic	Uses a simulated human; ENN reward head is small (frozen torso); single-turn only
Reproducibility	Code, model weights, and data publicly released on GitHub	No code; reproducibility relies on cited primary sources	Algorithms fully specified; depends on proprietary Gemini models for exact replication

Methodological contrast. FuseLLM and the exploration paper are both fundamentally about *data efficiency*, but at opposite ends of the LLM pipeline. FuseLLM extracts more signal from a fixed pretraining corpus by enriching one-hot token labels with multi-model probability distributions — the authors find that this yields the performance of standard continual training using 3.9× fewer tokens. The exploration paper extracts more signal from a fixed alignment budget by choosing *which* queries to label rather than relying on luck. The survey paper provides the context that explains why both efficiencies matter: pretraining cost is now astronomical, and RLHF datasets are the primary bottleneck in deploying truly well-behaved assistants.

4. Insights and Reflection

4.1 Trends emerging across the three papers

First, all three works treat **uncertainty and diversity as information**. FuseLLM uses the disagreement between source model distributions as a richer training signal than any single model's predictions; the exploration paper uses the variance of an ENN ensemble to identify queries whose answers will most reduce ambiguity; and the survey explicitly highlights ensembling, mixture-of-experts, and uncertainty-aware decoding as recurring themes in recent LLM design. The pattern suggests that the field is moving past point-estimate thinking toward *distributional* reasoning about model behaviour.

Second, there is a clear shift from **scaling alone** toward **better use of what already exists**. FuseLLM is explicit about this: rather than training a new model from scratch, it combines three existing ones. The survey documents the rise of LoRA, quantisation, distillation, and retrieval-augmented generation as ways to extract more capability from smaller footprints. The exploration paper makes the analogous argument for human feedback: rather than collecting

more labels, collect better ones.

Third, all three implicitly endorse the view that **fine-tuning and alignment matter as much as pretraining**. The survey devotes substantial space to instruction tuning, RLHF, DPO, constitutional AI, and red-teaming. FuseLLM positions itself partly as an alternative to distillation for transferring capability. The exploration paper directly studies the alignment stage. The era when pretraining accounted for nearly all of a model's quality appears to be ending.

4.2 Most promising directions

The double Thompson sampling result is striking because the gain compounds with scale: the authors show that the curve of queries-needed-by-passive versus queries-needed-by-double-TS is concave, meaning the relative advantage *grows* with the volume of feedback. If this extrapolates to industrial-scale RLHF (millions or billions of interactions), efficient exploration could plausibly shorten timelines to superhuman alignment by years. Knowledge fusion is similarly promising as a cost-amortisation strategy across organisations: rather than every lab retraining from scratch, capability could be pooled. The survey, meanwhile, highlights LLM-powered agents, multi-modal integration, and retrieval augmentation as the most active near-term frontiers.

4.3 Common limitations and open challenges

- **Hallucination and factual grounding.** All three papers acknowledge that LLMs continue to produce confident-sounding but false statements. None of the techniques studied directly solves this; FuseLLM can propagate the hallucinations of source models, and reward models in RLHF can be miscalibrated about truthfulness.
- **Evaluation is fragmented.** Each paper uses a different benchmark suite (BBH+CS+ME, the MMLU/SuperGLUE/HumanEval ecosystem, and win-rate against a baseline LLM). The lack of a shared evaluation makes cross-paper claims hard to verify.
- **Reliance on proprietary models.** The exploration paper depends on Gemini Nano and Pro; the survey catalogs many closed models (GPT-4, Gemini, Claude) whose architectures are not public. This limits reproducibility and concentrates research capacity in well-funded labs.
- **Single-turn assumption.** Both the exploration paper and most fine-tuning recipes described in the survey treat queries as independent. Real deployments are multi-turn, where earlier responses constrain later ones — an area the exploration paper itself flags as open.
- **Safety, bias, and privacy.** The survey devotes a full section to these and they remain open in both other papers. A fused model inherits the failure modes of its sources; an exploration agent that maximises informativeness may probe in directions humans would prefer it not to.

4.4 Future research directions

Three concrete directions stand out. (1) **Fusion of aligned and instruction-tuned models**, rather than just base pretrained ones — this would combine the FuseLLM and RLHF strands and could potentially fuse the safety behaviours of different alignment recipes. (2) **Exploration-aware data curation for pretraining**, applying the lessons of double Thompson sampling not just to preference queries but to the choice of pretraining documents. (3) **Multi-turn and deep exploration**, extending uncertainty-driven query selection to dialogue trajectories where the reward of a response depends on how it shapes future turns — the exploration authors explicitly call this out as the most important open problem in their concluding remarks.

5. Conclusion

Read together, these three papers tell a consistent story about where LLM research is heading. The era of capability-through-scale alone is giving way to an era of capability-through-better-use: better use of existing pretrained models (FuseLLM), better use of the techniques the community has already invented (the comprehensive survey), and better use of every human label collected during alignment (efficient exploration). The mathematical machinery driving these improvements — probability distributions over token vocabularies, ensemble-based uncertainty, Thompson sampling — is in many cases decades old; what is new is the recognition that the bottleneck in LLM development has shifted from raw compute toward judicious decisions about *what* to train on and *which* questions to ask.

For practitioners, the immediate takeaway is that several order-of-magnitude efficiency gains remain on the table without requiring new hardware or new model architectures. For researchers, the open frontiers — multi-turn exploration, fusion of aligned models, robust uncertainty quantification at the trillion-parameter scale — are clearly identified and tractable. And for the broader field, the survey's reminder that hallucination, bias, and safety remain unsolved is a useful counterweight to the optimism of the methodological papers: efficiency gains in capability do not automatically translate into efficiency gains in trustworthiness, and the latter may be the more important target for the

next phase of LLM research.